

Linux Administration

Regular Expressions

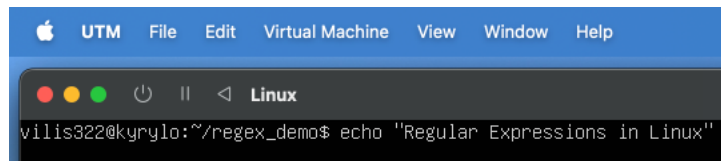
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Introduction: What Are Regular Expressions?

Regular expressions (regex) are text patterns used to search, match, and manipulate strings.

They are powerful tools for system administrations, developers, and data analysts. In Linux, regex are used inside many commands such as **grep**, **sed**, and **awk**.

A screenshot of a Linux terminal window. The window has a blue title bar with the text 'UTM File Edit Virtual Machine View Window Help'. Below the title bar is a dark grey bar with the text 'Linux'. The terminal shows a prompt 'vilis322@krylo:~/regex_demo\$' followed by the command 'echo "Regular Expressions in Linux"'.

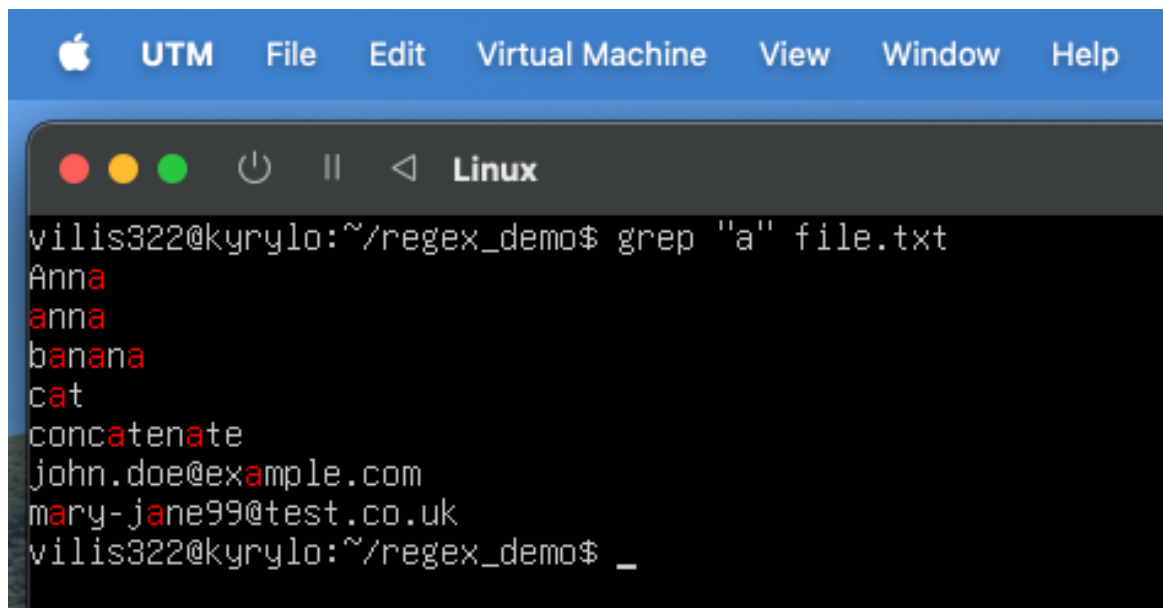
```
vilis322@krylo:~/regex_demo$ echo "Regular Expressions in Linux"
```

Basic Concept and Metacharacters

The simplest way to use a regular expression is to search for a specific character or word.

The **grep** command scans each line of a file and prints those that match the given pattern.

Common metacharacters include `.`, `^`, `$`, `*`, `+`, `?`, `[]`, and `{}`.



```
UTM  File  Edit  Virtual Machine  View  Window  Help

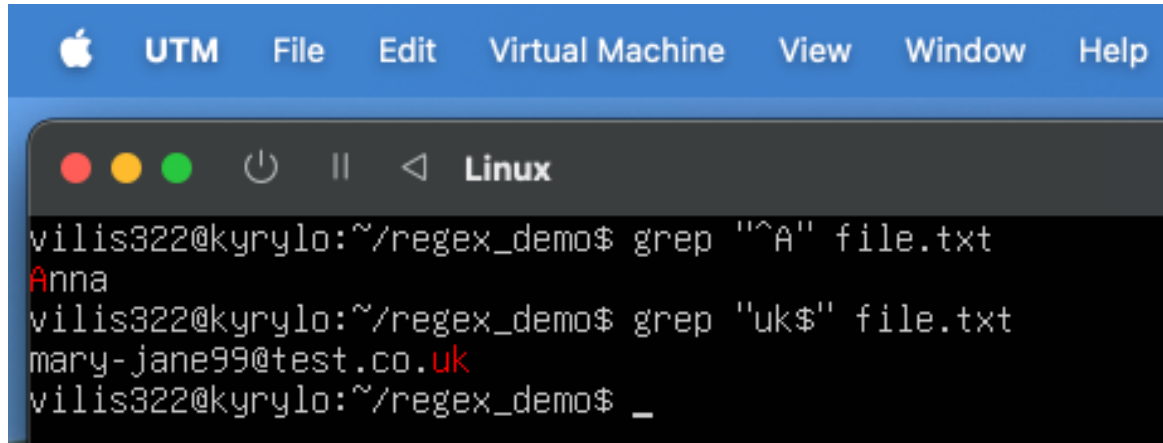
Linux

vilis322@kyrylo:~/regex_demo$ grep "a" file.txt
Anna
anna
banana
cat
concatenate
john.doe@example.com
mary-jane99@test.co.uk
vilis322@kyrylo:~/regex_demo$ _
```

I used **grep** to find all lines in file.txt that contain the letter “a”.

Anchors: Beginning and End of Line

Anchors are used to match positions rather than characters. **^** matches the beginning of a line, while **\$** matches the end. They are useful for validating formats like headers or endings in logs.

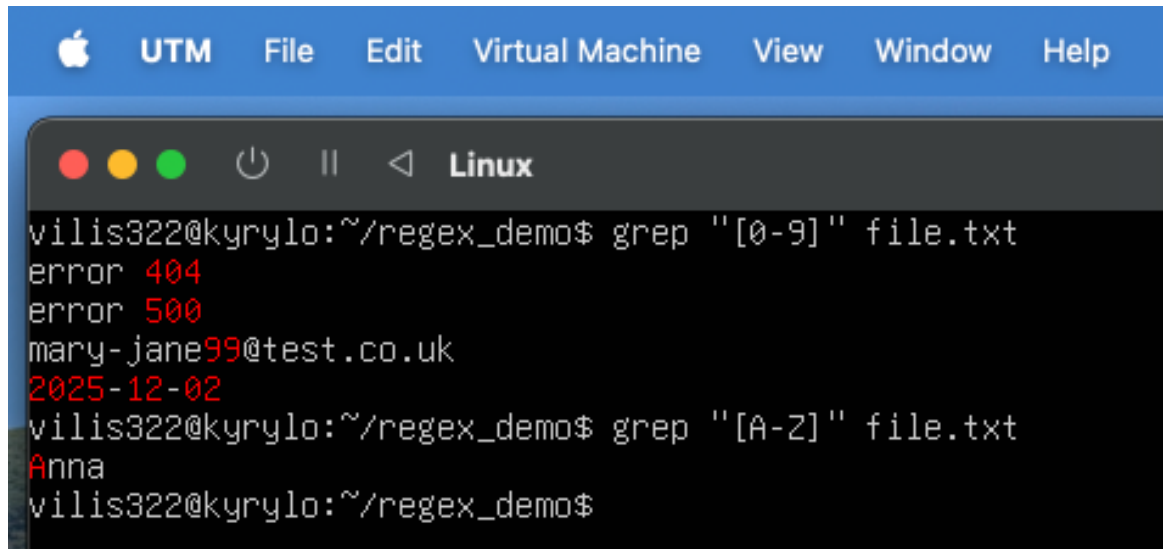
A screenshot of a UTM virtual machine window. The title bar is blue with the UTM logo and menu items: UTM, File, Edit, Virtual Machine, View, Window, Help. Below the title bar is a dark grey bar with window controls (red, yellow, green buttons, a power button, a pause button, and a back button) and the text "Linux". The main area is a black terminal window with white text. It shows the following commands and output:

```
vilis322@kyrylo:~/regex_demo$ grep "^A" file.txt
Anna
vilis322@kyrylo:~/regex_demo$ grep "uk$" file.txt
mary-jane99@test.co.uk
vilis322@kyrylo:~/regex_demo$ _
```

The first command finds lines starting with **"A"**, and the second finds lines that end with **"uk"**.

Character Classes

Character classes **[]** define a set of range of allowed characters. For example, **[0-9]** matches any digit, and **[A-Z]** matches any uppercase letter. Also can combine or negate classes, for example, **[^a-z]** means “any character except lowercase letters”.



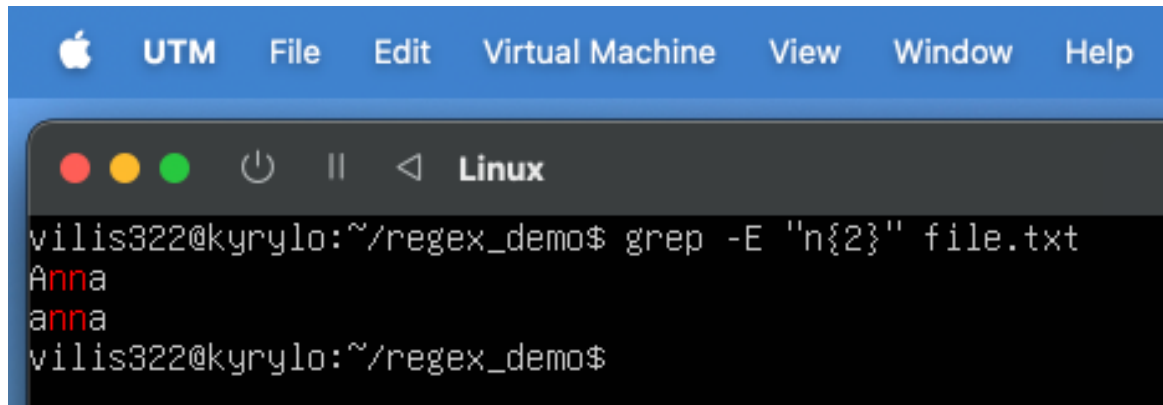
```
UTM  File  Edit  Virtual Machine  View  Window  Help
Linux
vilis322@krylo:~/regex_demo$ grep "[0-9]" file.txt
error 404
error 500
mary-jane99@test.co.uk
2025-12-02
vilis322@krylo:~/regex_demo$ grep "[A-Z]" file.txt
Anna
vilis322@krylo:~/regex_demo$
```

I searched for lines containing digits and then uppercase letters.

Quantifiers (Repeating Patterns)

Quantifiers specify how many times a character or group may appear. `*` = 0 or more, `+` = 1 or more, `{n, m}` = between n and m times.

They are often used to find repeating sequences or validate input length.



```
vilis322@kyrylo:~/regex_demo$ grep -E "n{2}" file.txt
Anna
anna
vilis322@kyrylo:~/regex_demo$
```

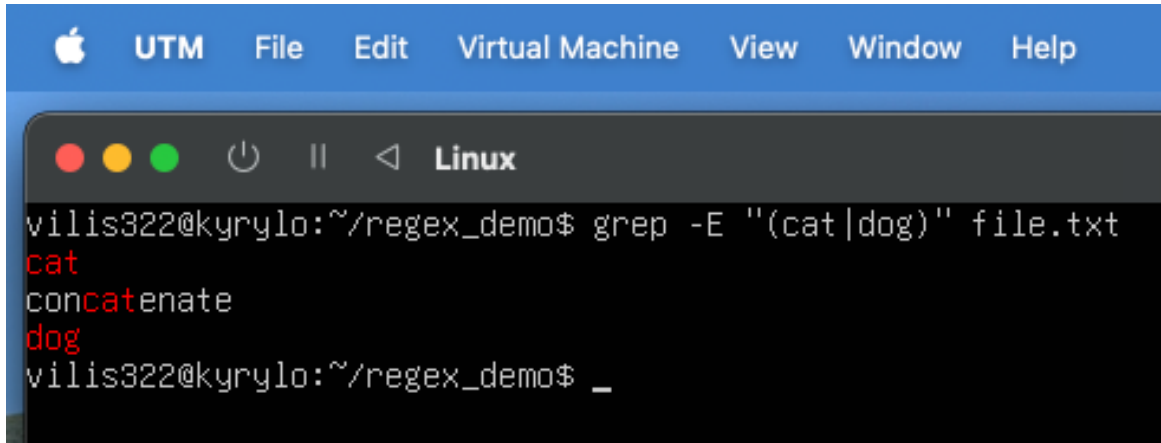
This command finds all lines where the letter “**a**” appears two times in a row.

Alternation and Grouping

Alternation `|` means “or”.

Parentheses `()` group expressions together.

Combined, they let you search for several options within one pattern.



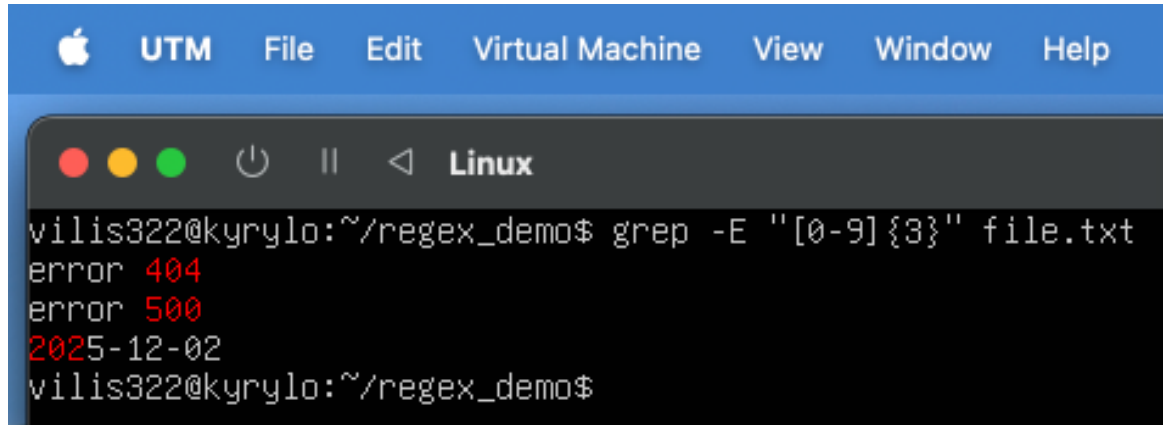
```
UTM  File  Edit  Virtual Machine  View  Window  Help
Linux
vilis322@kyrylo:~/regex_demo$ grep -E "(cat|dog)" file.txt
cat
concatenate
dog
vilis322@kyrylo:~/regex_demo$ _
```

I searched for lines that contain either “cat” or “dog”.

Extended Regular Expressions

Extended Regular Expressions allow additional syntax such as **+**, **?**, **{}**, and **|** without backslashes.

In Linux, it's enable this mode with the **-E** flag or by using **egrep**.



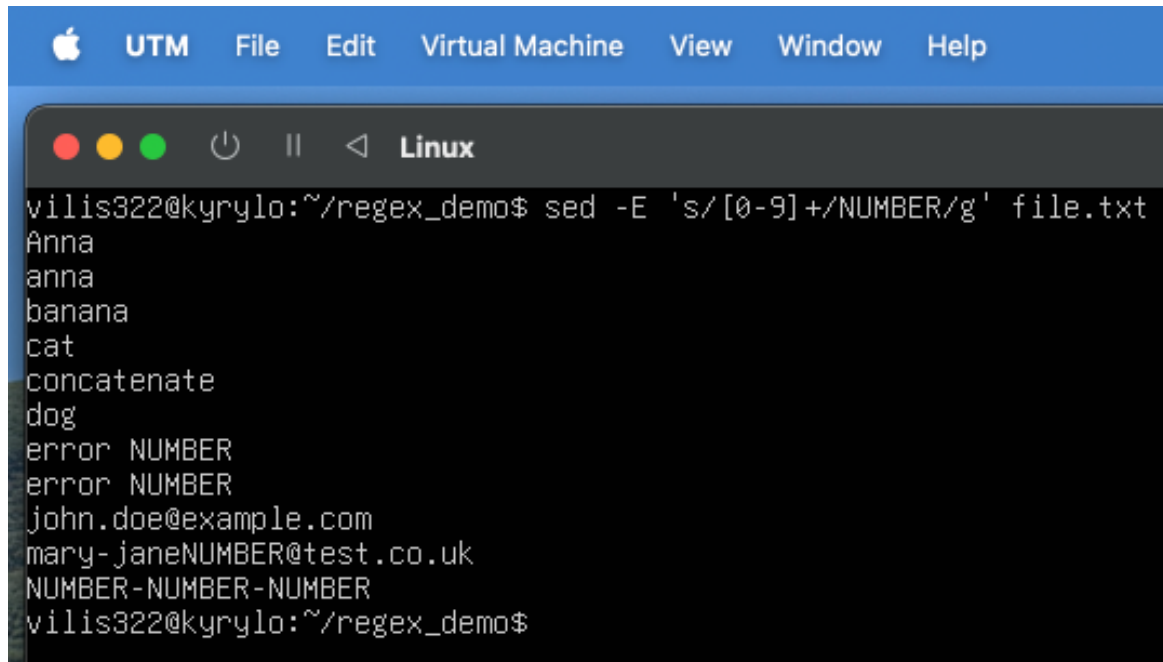
```
vilis322@kyrylo:~/regex_demo$ grep -E "[0-9]{3}" file.txt
error 404
error 500
2025-12-02
vilis322@kyrylo:~/regex_demo$
```

I used **grep -E** to find lines that include three consecutive digits.

Using Regex with 'sed'

sed is a stream editor that can replace text using regex.

The expression **s/pattern/replacement/g** means "substitute every match of pattern with replacement".



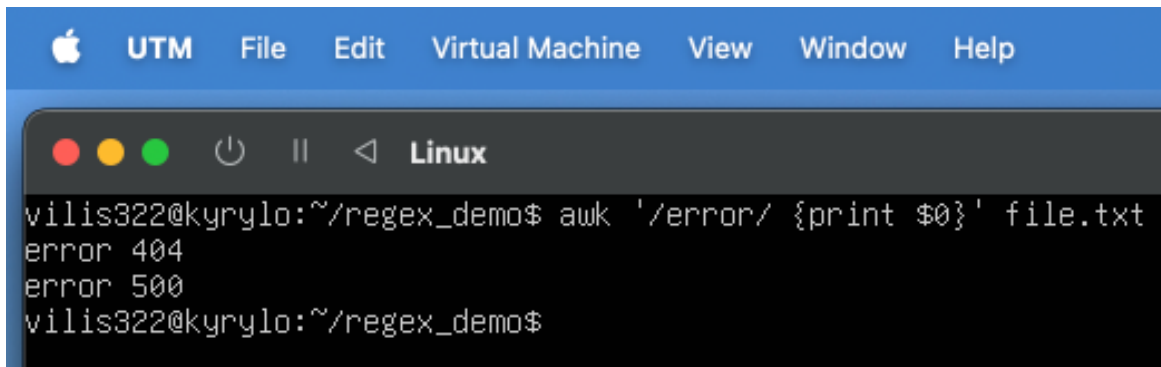
```
UTM File Edit Virtual Machine View Window Help
Linux
vilis322@kyrylo:~/regex_demo$ sed -E 's/[0-9]+/NUMBER/g' file.txt
Anna
anna
banana
cat
concatenate
dog
error NUMBER
error NUMBER
john.doe@example.com
mary-janeNUMBER@test.co.uk
NUMBER-NUMBER-NUMBER
vilis322@kyrylo:~/regex_demo$
```

I replaced all numeric sequences in the file with the word NUMBER.

Using Regex with 'awk'

awk processes text line by line and executes actions when patterns match.

Here, the pattern **/error/** selects lines containing the word "error", and **{print \$0}** prints the entire matching line.



```
UTM  File  Edit  Virtual Machine  View  Window  Help

Linux

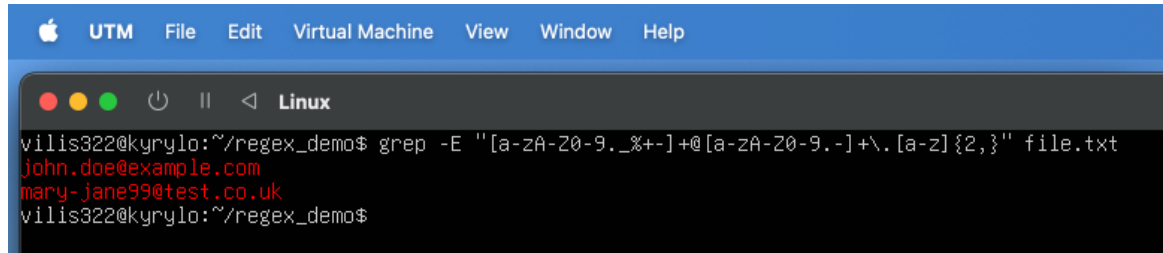
vilis322@krylo:~/regex_demo$ awk '/error/' {print $0}' file.txt
error 404
error 500
vilis322@krylo:~/regex_demo$
```

Practical Example: Email Search

Regex can validate complex formats such as email addresses.

This expression looks for text that has the general email structure

username@domain.extension.



```
UTM File Edit Virtual Machine View Window Help
Linux
vilis322@kynylo:~/regex_demo$ grep -E "[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-z]{2,}" file.txt
john.doe@example.com
mary-jane99@test.co.uk
vilis322@kynylo:~/regex_demo$
```

Conclusion

Regular expressions are essential tools in Linux for text analysis, validation, and automation.

They save time and simplify tasks like log inspection, data extraction, and configuration parsing.

Understanding regex improves both productivity and precision in system administration.